



# Morality & Ethics in Science

# Moral Questions in Science

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- May I perform experiments that might harm human subjects?
- Must I reject data if I suspect it to be falsified? Must I report a colleague whom I suspect of such falsification?
- Who should be counted as an author of a scientific manuscript?
- What research topic should I choose?

# Moral Answers

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## Permission

e.g. "You ***may*** perform this experiment"

## Obligation

e.g. "You ***should*** count your collaborator as a co-author"

## Prohibition

e.g. "You ***must not*** falsify data".

# What is Morality?

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## Descriptive view

A code of conduct that is put forward by a society, group, or individual, that is accepted as a guide to behavior by its members, or by that individual.

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Distinguish morality from:

- Convention
- Etiquette
- Law
- (Self-interested) instrumental rationality

# What is Morality?

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A code of conduct that is put forward by a society, group, or individual, that is accepted as a guide to behavior by its members, or by that individual.

Examples of scientific codes of conduct:

- Nuremberg Code of legitimate medical experiments (1949)
- Declaration of Helsinki on human experimentation (1975)
- Swedish Association of Graduate Engineers (Sveriges ingenjörer) – Honor Code for Engineers, <http://www.sverigesingenjorer.se/Om-forbundet/Sa-tycker-vi/hederskodex/> (in Swedish)
- APS (American Physical Society) Guidelines for Professional Conduct, [http://www.aps.org/policy/statements/02\\_2.cfm](http://www.aps.org/policy/statements/02_2.cfm)

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## Normative view:

The code of conduct that all rational persons, under certain specified conditions, would endorse.

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## Research Ethics:

- Why should I endorse a certain code of behavior?
- Why is it rational for all scientists to endorse certain codes of behavior?

# Ethics vs. Morality

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## Ethics:

A theory that offers normatively valid reasons for rationally endorsing a code of behavior

## Research Ethics:

- Why should I endorse a certain code of behavior?
- Why is it rational for all scientists to endorse certain codes of behavior?
- Adds to reasons for method choice

| Method A   |            |
|------------|------------|
| <i>pro</i> | <i>con</i> |
| ...        | ...        |

| Method B   |            |
|------------|------------|
| <i>pro</i> | <i>con</i> |
| ...        | ...        |



# Ethics vs. Morality

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## Ethics:

A theory that offers normatively valid reasons for rationally endorsing a code of behavior

Even if we agree on a code of conduct, we might not agree on why we agree on it.

You are currently developing a safety-increasing technology  $L$  for lorries, which is expected to save 10 lives in the first year of its introduction. Now you learn that a different technology  $C$  built into cars, which could be developed from your current results in your lab, would save 100 lives. Your lab is the only one with the capacity to develop one of these technologies (but not both) within this year. Which do you choose?

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Technology C

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Please pause the video and write down your answer before continuing

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  - Cannot realise both at the same time
  - Principle: saving more lives is better than saving less
- => Realize technology *C*

You run a hospital that has 100 patients dying of different organ failures per year. There are 10 healthy patients whose organs could be taken (which would imply their death) and transplanted to save these 100 patients. There are no relevant consequences other than the saving of the 100 and the death of the 10 to be considered. What do you do?

(T) Transplant      (D) Don't transplant

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- Don't transplant – saves 10 lives
  - Transplant – saves 100 lives
  - Cannot realise both at the same time
  - ~~• Principle: saving more lives is better than saving less~~
  - Principle: don't directly cause the death of people
- => Don't transplant



# Different Ethical Intuitions

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To behave morally might mean:

1. Bringing about the best ***consequences***
2. Following relevant ***rules*** and ***duties***



# Different Ethical Intuitions

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To behave morally might mean:

1. Bringing about the best ***consequences***
2. Following relevant ***rules*** and ***duties***
3. Exemplifying relevant ***character traits*** or ***virtues***

Three frameworks for normative ethics

Part I: Consequentialism & Deontology

**Consequentialism:** Choices are to be morally assessed only by their consequences

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**Deontology:** To choose morally is to fulfill relevant rules and duties

**Virtue ethics:** Morality consists in having and exemplifying good character traits

# Consequentialism

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Choices are to be morally assessed only by their consequences

# Consequentialism

---

Choices are to be morally assessed only by their consequences

- How to evaluate consequences?
  - e.g, utilitarians: pleasure, happiness, desire satisfaction, or “welfare”

Morality consists in “the rules and precepts for human conduct, by the observance of which [a happy existence] might be, to the greatest extent possible, secured.” (J. S. Mill 1861)





## Relevant for scientists:

- Consequences for the epistemic goals of science (good predictions, explanations,...)
- Consequences for the institution of science (Impartiality, trust,...)
- Consequences for the use of scientific results (technological progress, new risk, policies,...)

- **Problem 1:** “extreme demandingness”: there is no realm of moral permissions, no realm of going beyond one's moral duty (supererogation), no realm of moral indifference. All acts are seemingly either required or forbidden.

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- **Problem 2:** permitting intuitively wrong acts – e.g. killing one for the benefit of many

# Deontology

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- “deontology” – from Greek *deon* (duty)

# Deontology

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To choose morally is to fulfill relevant rules and duties

- “deontology” – from Greek *deon* (duty)
- some choices cannot be justified by their effects—no matter how good their consequences, some choices are forbidden and others mandated

People have certain rights, which must be respected

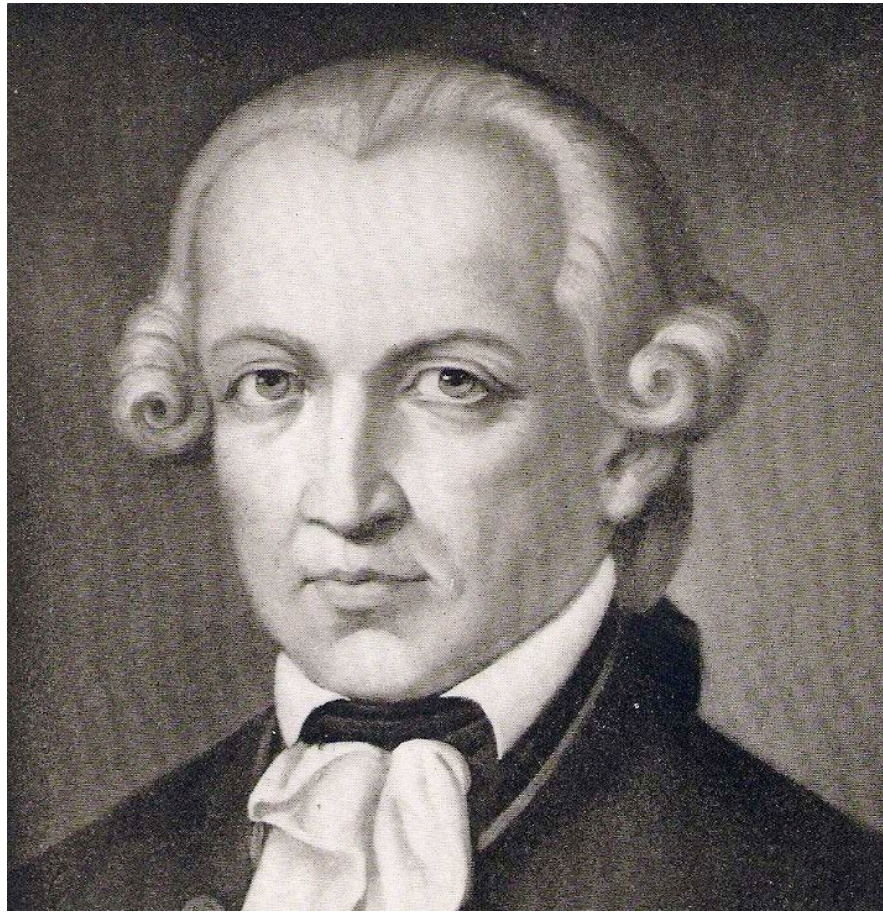
People have certain rights, which must be respected

Certain intentions are not permissible (e.g. intending to harm or deceive)

## **Categorical Imperative (I. Kant 1785):**

“Act only according to that maxim whereby you can, at the same time, will that it should become a universal law.”





## Relevant for scientists, e.g.:

- Treating humans as autonomous ends, not as instruments for some other purpose
- Prohibition to intend or cause harm, irrespective of other benefits
- Speaking truthfully & defending the truth
- Intellectual property rights

Three frameworks for normative ethics

Part II: Deontology & Virtue Ethics



# Advantages over consequentialism



## Advantages over consequentialism:

- Use of checklist – no need to consider all possible consequences

## Examples of such checklists:

- Nuremberg Code of legitimate medical experiments (1947)
- Declaration of Helsinki on human experimentation (1975)
- Swedish Association of Graduate Engineers (Sveriges ingenjörer) – Honor Code for Engineers, <http://www.sverigesingenjorer.se/Om-forbundet/Sa-tycker-vi/hederskodex/> (in Swedish)
- APS (American Physical Society) Guidelines for Professional Conduct, [http://www.aps.org/policy/statements/02\\_2.cfm](http://www.aps.org/policy/statements/02_2.cfm)



## Advantages over consequentialism:

- For the deontologist, there are acts that are neither morally wrong demanded, or praiseworthy.



## Advantages over consequentialism:

- For the deontologist, there are acts that are neither morally wrong demanded, or praiseworthy.
- Leaves space for agents to give special concern to *their* projects



- **Problem 1:** compliance with deontological norms might bring about disastrous consequences



- **Problem 1:** compliance with deontological norms might bring about disastrous consequences.
- **Problem 2:** all rights and duties are categorical. no systematic way of distinguishing degrees of wrongness



# Virtue Ethics

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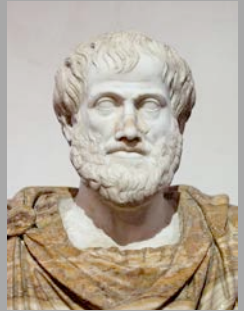
Virtue ethics: Morality consists in having and exemplifying good character traits (i.e. “virtues”)



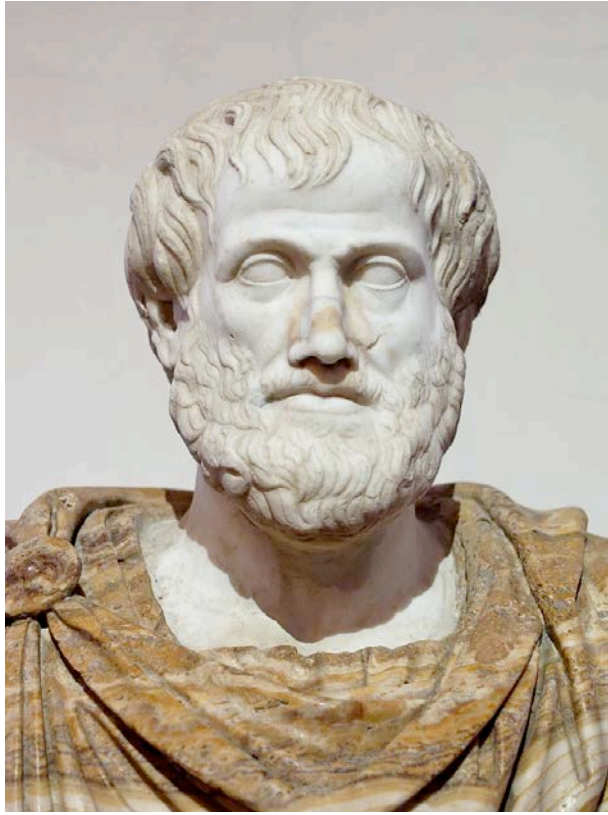
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Aristotle  
384–322 BC





## Relevant virtues for scientists, e.g.:

- **courage** (standing up at the appropriate time for what one believes in despite some potential personal cost)
- **respectfulness** (treating others with the respect they deserve)
- **resoluteness** (staying with one's work, forging on despite difficulties, within the bounds of reason)
- **sincerity** (being honest and truthful when appropriate, believing what you say)
- **humility** (giving due weight to one's strengths and weaknesses)
- **reflexivity** (being critical enough of one's work, making due allowances for one's own biases)



## Advantages over consequentialism

- avoids “overdemandingness”

## Advantages over deontology

- explains what motivates people to behave morally



- **Problem 1:** Doesn't give clear guidance how to behave in particular situations
- **Problem 2:** Difficult to resolve moral conflicts

# Summary

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**Consequentialism:** Choices are to be morally assessed solely by their consequences

**Deontology:** To choose morally is to fulfill relevant rules and duties

**Virtue ethics:** Morality consists in having and developing good character traits

# Morally Permissible Experiments



# Morally Permissible Experiments

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“Tuskegee Study of Untreated Syphilis in the  
Negro Male”  
U.S. Public Health Service, 1932-1972



# Morally Permissible Experiments

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- Deceived about length of trial

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# Morally Permissible Experiments

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# Morally Permissible Experiments

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- Deceived about length of trial
- Not informed about kind of disease
- Not informed about potential cures
- Effective treatments withheld

“Tuskegee Study of Untreated Syphilis in the  
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# Informed Consent

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- Informed, voluntary, and decisionally-capacitated consent

- **Informed:** a research subject is informed if she has received all relevant information about the purpose of the research project, how it will be carried out, and in what way it will affect her.

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- **Decisionally-capacitated consent:** the subject has decisional capacity to assess the information provided, appreciate in what way it concerns her, as well as make and communicate a decision in favor of participating.

# Why Informed Consent?

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|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Consequentialism                                                                                                                                |
| <ul style="list-style-type: none"><li>• individuals know best what is good for them</li><li>• establish and maintain trust in science</li></ul> |

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| Consequentialism                                                                                                                                | Deontology                                                                                                                          | Virtue ethics                                                                                              |
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# Morally Permissible Experiments

## The long read The cult of Mary Beard Charlotte Higgins Journal

# The Guardian

### German ambassador tells Brexiters: stop fixating on second world war

By Rachel Tuck in Germany  
The German ambassador to the UK, Peter Ahlborn, has told a group of Brexiters that they should stop fixating on the Second World War and instead focus on the future of the country.

### VW tested diesel fumes on monkeys and humans



### Hadley Freeman What the reverence for Roman Polanski tells us about Hollywood hypocrisy



abc NEWS

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SEARCH

## German automakers face heat over tests on monkeys, humans

By DAVID MCHUGH AND GEIR MOULSON, ASSOCIATED PRESS  
FRANKFURT, Germany — Jan 29, 2016, 1:57 PM ET



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L'actualité internationale 24h / 24

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LA CHAÎNE EN DIRECT

07:51

(Heure de Paris) REVUE DE PRESSE

Economie

Volkswagen | BMW | Dieselgate

## Diesel : Volkswagen, Daimler et BMW accusés d'avoir mené des tests sur des humains et des singes

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## Volkswagen supervisory board demands inquiry into diesel fume tests on monkeys and people

Updated about 11 hours ago

The German Government has denounced tests funded by the country's carmakers, in which humans and monkeys reportedly inhaled diesel exhaust fumes.

Carmakers Volkswagen, Daimler and BMW funded a research organisation which reportedly conducted the testing on 10 monkeys and 25





# Morally Permissible Experiments

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## The Diesel Experiment Case

- RWTH Aachen, Centre for workplace and environmental medicine
- Consenting subjects were exposed to 1.5 ppm NO<sub>2</sub> for 3 hours
- Until 2002: accepted daily expose to NO<sub>2</sub> 5 ppm. Since 2009: 0.5 ppm
- Testing “light symptoms” (e.g. rashes), blood test, inflammation index, lung function



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**Was this experiment morally permissible?**

Please pause the video and write down your answer before continuing



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- Informed consent rule ✓
- No excessively risky consequences ✓
- Sincerity of researchers motivation ✗

# Dealing With Misconduct



# Scientific Incompetence vs. Misconduct

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# Scientific Misconduct

## Jan-Hendrik Schön Case

### Bell Labs launches inquiry into allegations of data duplication

**Small-Scale Data Duplication**

The electronic band-gap semiconductors system following data duplication in some of its papers, says Bell Labs. The findings, which are expected to be published in the journal *Science*, are the result of an investigation by a team of scientists at Bell Labs. The findings are expected to be published in the journal *Science*.



### Misconduct finding at Bell Labs shakes physics community

**Scientific misconduct**

The finding of misconduct at Bell Labs has shaken the physics community. The finding is expected to be published in the journal *Science*.

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Jan-Hendrik Schön, Bell Labs physicist, is central to the data duplication case.

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Bell Labs building, the site of the data duplication case.

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### Bell Labs inquiry spreads to superconductors

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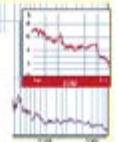
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### commentary



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### Is a bell tolling for Bell Labs?

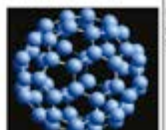
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# Why is Scientific Misconduct not Admissible?

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|                                                                                               |
|-----------------------------------------------------------------------------------------------|
| <b>Consequentialism</b>                                                                       |
| <ul style="list-style-type: none"><li>• Undermines trust</li><li>• Wastes resources</li></ul> |

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| Consequentialism                                                                              | Deontology |
|-----------------------------------------------------------------------------------------------|------------|
| <ul style="list-style-type: none"><li>• Undermines trust</li><li>• Wastes resources</li></ul> | Don't lie! |



# Why is Scientific Misconduct not Admissible?

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| Consequentialism                                                                              | Deontology | Virtue Ethics                                                                  |
|-----------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Undermines trust</li><li>• Wastes resources</li></ul> | Don't lie! | <ul style="list-style-type: none"><li>• Sincerity</li><li>• Humility</li></ul> |

# Scientific Misconduct – Main Instances

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- **Fabrication:** making up data or results.
- **Falsification:** manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research record.

# Scientific Misconduct – Main Instances

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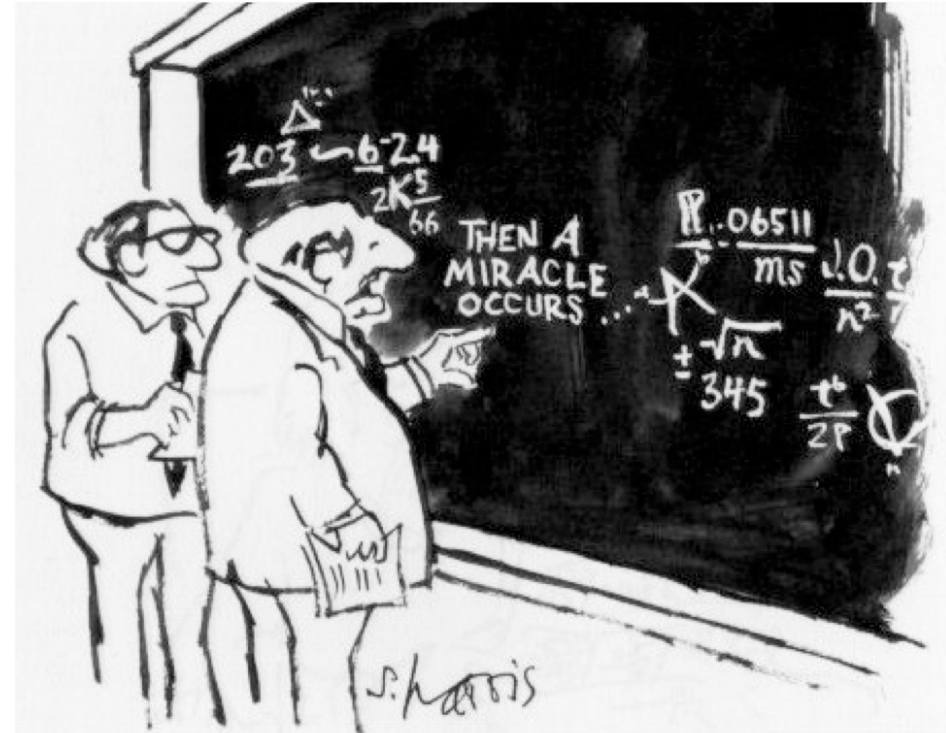
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- **Falsification:** manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research record.
- **Plagiarism:** the appropriation of another person's ideas, processes, results, or words without giving appropriate credit.

# Scientific Misconduct – Difficult Cases

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# Scientific Misconduct – Difficult Cases

- Black-boxed science

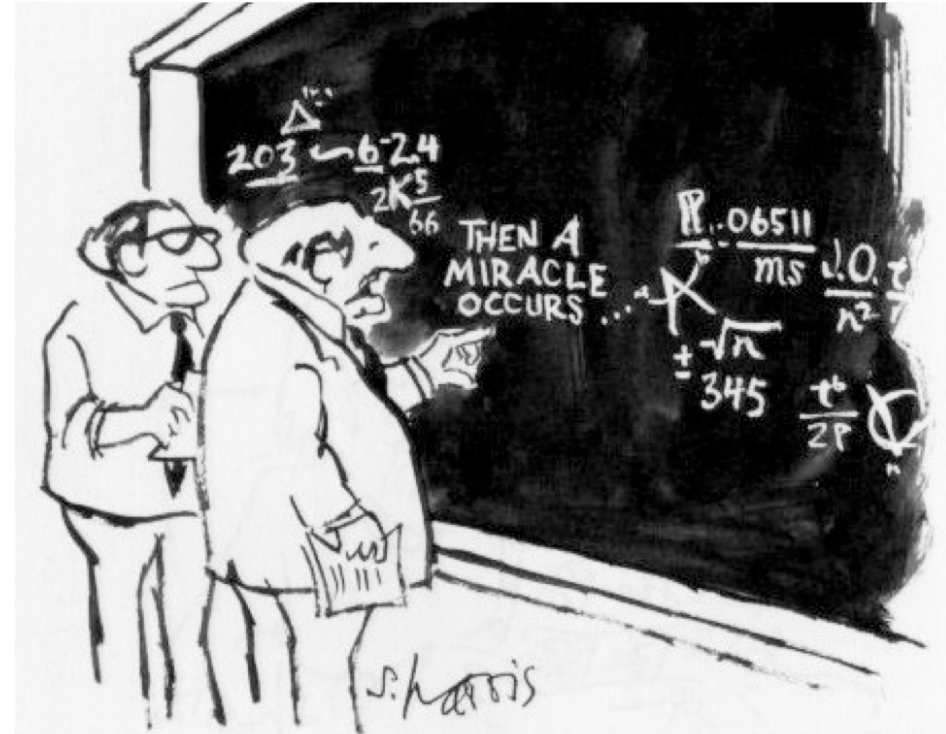


"I think you should be more explicit here in step two."

from *What's so Funny about Science?* by Sidney Harris (1977)

# Scientific Misconduct – Difficult Cases

- Black-boxed science
- Importance of peer review

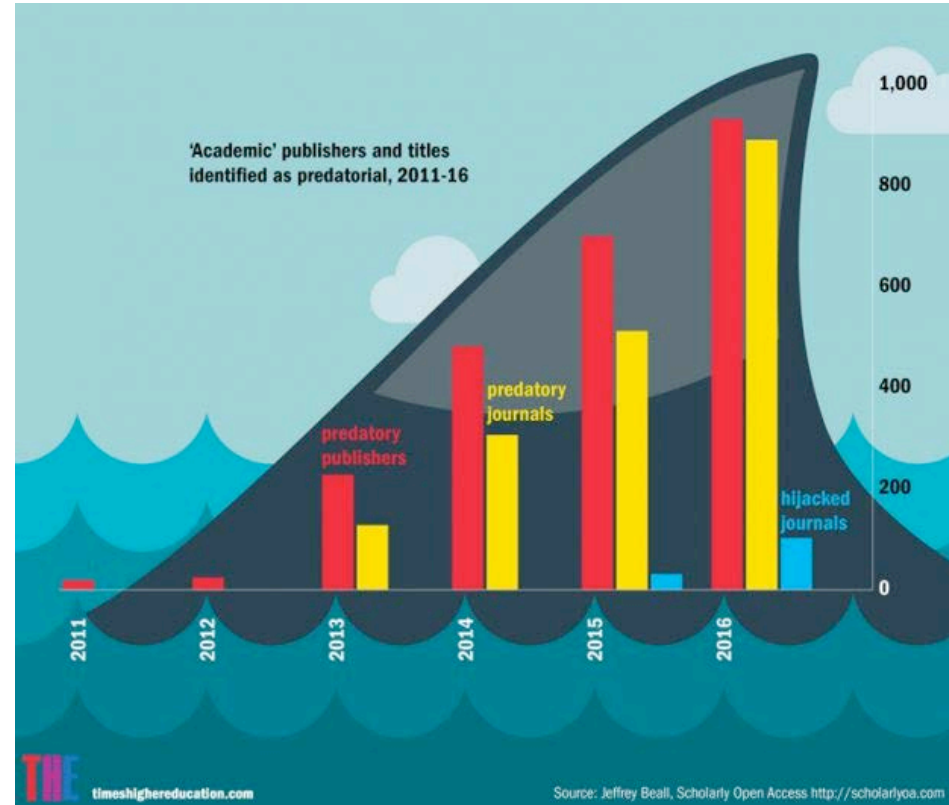


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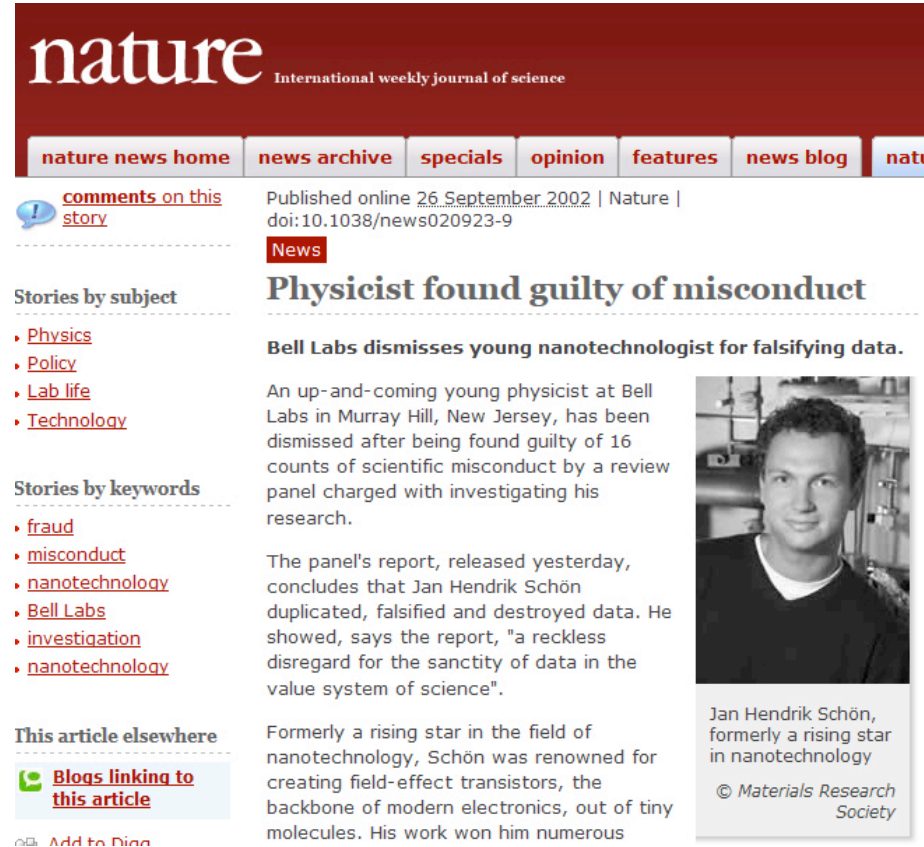
# Scientific Misconduct – Difficult Cases

- Black-boxed science
- Importance of peer review
- Threat from predatory publishers



# Scientific Misconduct – Difficult Cases

- Black-boxed science
- Importance of peer review
- Threat from predatory publishers



The screenshot shows the top of a Nature news article. The header includes the 'nature' logo and navigation links like 'nature news home', 'news archive', 'specials', 'opinion', 'features', 'news blog', and 'nat'. The article title is 'Physicist found guilty of misconduct' with a sub-headline 'Bell Labs dismisses young nanotechnologist for falsifying data.' The text describes how Jan Hendrik Schön was dismissed after being found guilty of falsifying data. A photo of Schön is on the right, with a caption identifying him as a former rising star in nanotechnology. The bottom of the page shows a 'This article elsewhere' section with a link to 'Blogs linking to this article' and a 'Add to Digg' button.

**nature** International weekly journal of science

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[comments on this story](#) Published online 26 September 2002 | Nature | doi:10.1038/news020923-9

**News**

## Physicist found guilty of misconduct

### Bell Labs dismisses young nanotechnologist for falsifying data.

An up-and-coming young physicist at Bell Labs in Murray Hill, New Jersey, has been dismissed after being found guilty of 16 counts of scientific misconduct by a review panel charged with investigating his research.

The panel's report, released yesterday, concludes that Jan Hendrik Schön duplicated, falsified and destroyed data. He showed, says the report, "a reckless disregard for the sanctity of data in the value system of science".

Formerly a rising star in the field of nanotechnology, Schön was renowned for creating field-effect transistors, the backbone of modern electronics, out of tiny molecules. His work won him numerous



Jan Hendrik Schön, formerly a rising star in nanotechnology

© Materials Research Society

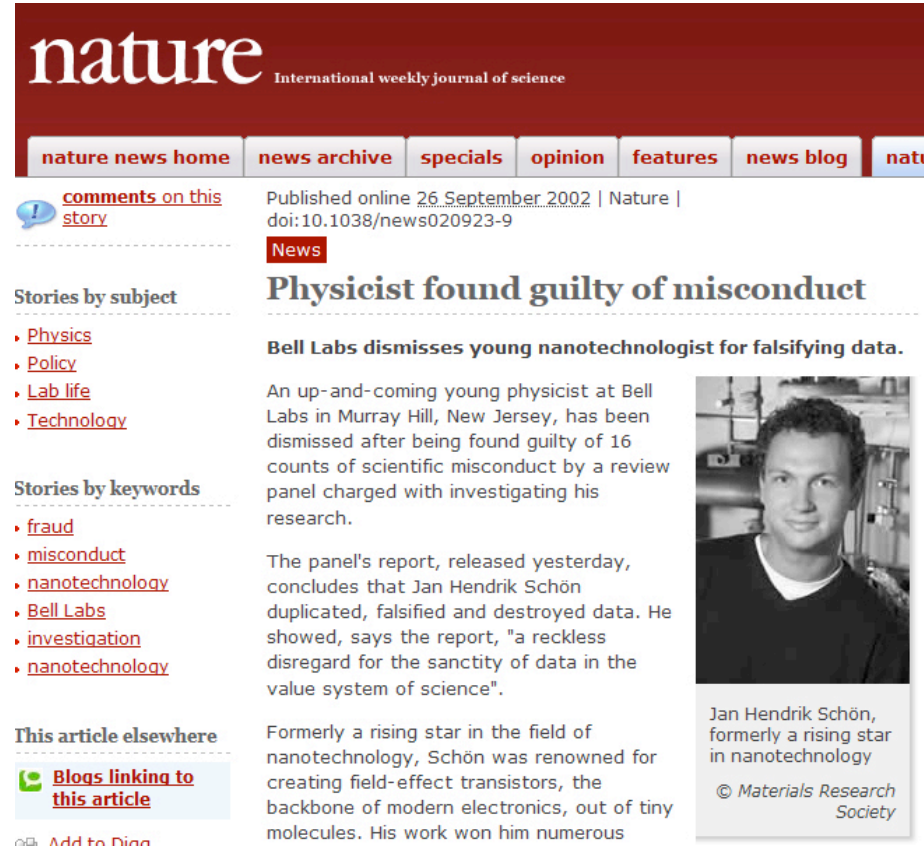
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# Scientific Misconduct – Difficult Cases

- Black-boxed science
- Importance of peer review
- Threat from predatory publishers
- Importance of transparency



The screenshot shows the top of a Nature news article. The header is dark red with the 'nature' logo in white. Below the header is a navigation bar with links: 'nature news home', 'news archive', 'specials', 'opinion', 'features', 'news blog', and 'nat'. The article title is 'Physicist found guilty of misconduct' in a large, bold, black font. Below the title is a sub-headline: 'Bell Labs dismisses young nanotechnologist for falsifying data.' The main text begins with 'An up-and-coming young physicist at Bell Labs in Murray Hill, New Jersey, has been dismissed after being found guilty of 16 counts of scientific misconduct by a review panel charged with investigating his research.' A black and white photo of Jan Hendrik Schön is on the right. Below the photo is a caption: 'Jan Hendrik Schön, formerly a rising star in nanotechnology' and '© Materials Research Society'. On the left side of the article, there are sections for 'Stories by subject' (Physics, Policy, Lab life, Technology) and 'Stories by keywords' (fraud, misconduct, nanotechnology, Bell Labs, investigation, nanotechnology). At the bottom left, there is a section 'This article elsewhere' with a link 'Blogs linking to this article' and a small icon of a book.

**nature** International weekly journal of science

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**News**

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Jan Hendrik Schön, formerly a rising star in nanotechnology

© Materials Research Society

**Stories by subject**

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**This article elsewhere**

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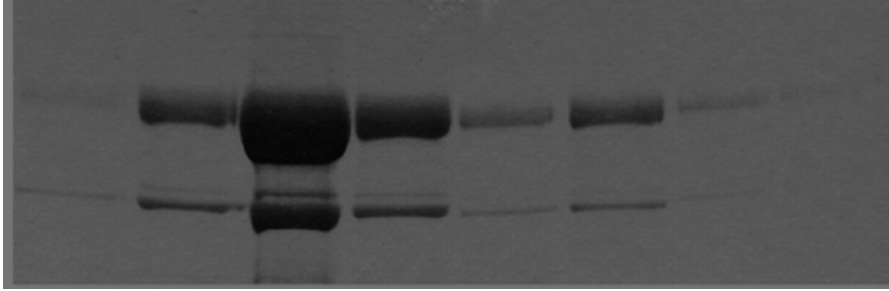
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# Scientific Misconduct – Difficult Cases

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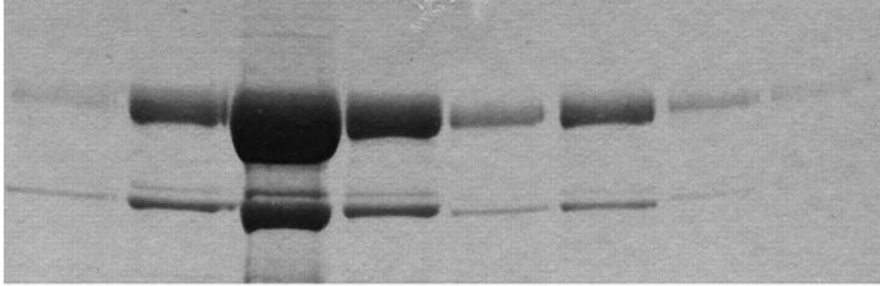
# Scientific Misconduct – Difficult Cases

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# Scientific Misconduct – Difficult Cases

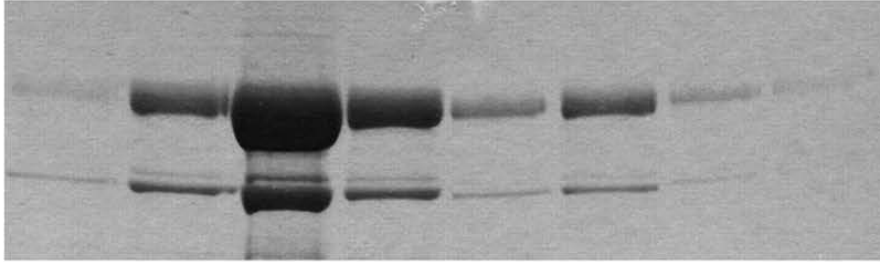
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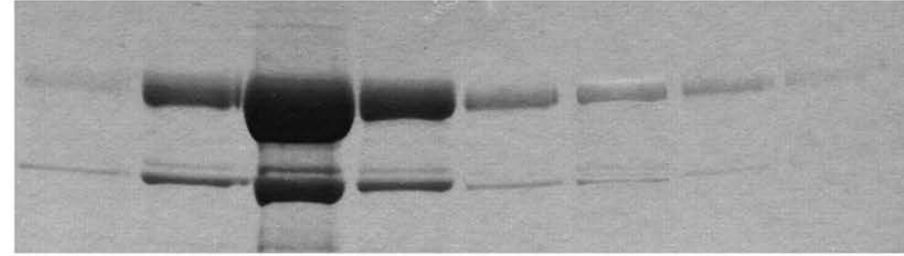
# Scientific Misconduct – Difficult Cases

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A. Original image



Manipulated image

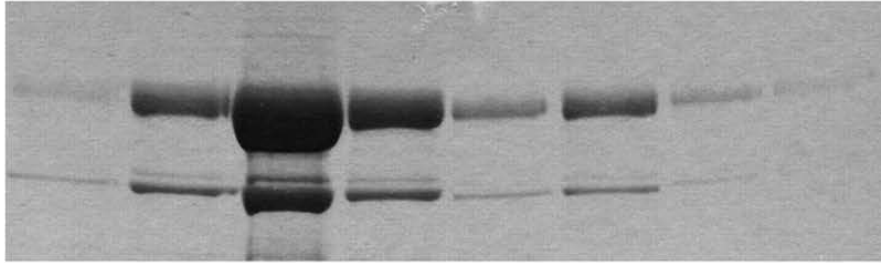




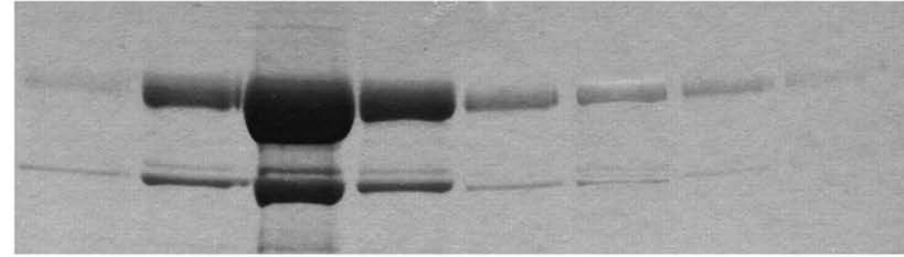
# Scientific Misconduct – Difficult Cases

A.

Original image



Manipulated image



Violates the guideline that “No specific feature within an image may be enhanced, obscured, moved, removed, or introduced.”



# Scientific Misconduct – Prevalence

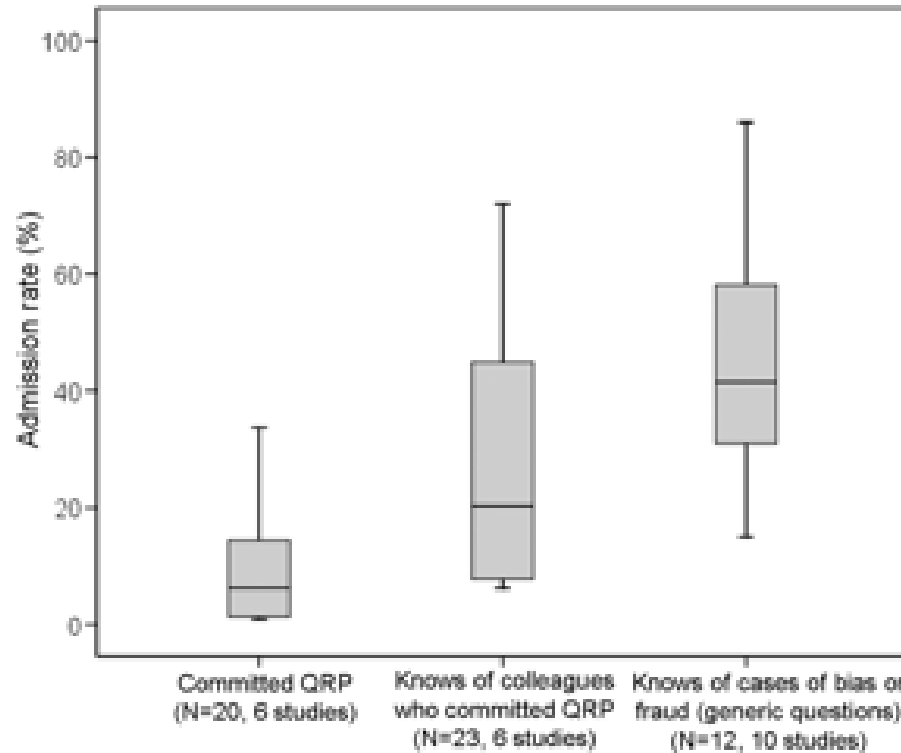


Figure 3. Admission rates of Questionable Research Practices (QRP) in self- and non-self-reports.



# Scientific Misconduct – Countermeasures

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- Responsibility of coauthors
- Responsibilities of research institutions
- Responsibilities of scientific colleagues
- Responsibility of journals



# Scientific Misconduct – Countermeasures

---

- Responsibility of coauthors
- Responsibilities of research institutions
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- Responsibility of journals

# Whistleblowing

---

Whistleblower: person who exposes any kind of information or activity that is deemed illegal, unethical, or not correct within an organization that is either private or public.

# Why Whistleblow?

---

Whistleblower: person who exposes any kind of information or activity that is deemed illegal, unethical, or not correct within an organization that is either private or public.

| Consequentialism                                                                              | Deontology |
|-----------------------------------------------------------------------------------------------|------------|
| <ul style="list-style-type: none"><li>• Undermines trust</li><li>• Wastes resources</li></ul> | Don't lie! |

# Moral Rules for Authorship



## Till Grüne-Yanoff

Professor of Philosophy, [Royal Institute of Technology \(KTH\)](#), Stockholm

Verified email at kth.se - [Homepage](#)

[Philosophy of Science](#) [Decision Theory](#)

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T Grüne-Yanoff  
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SO Hansson, T Grüne-Yanoff  
Stanford encyclopedia of philosophy

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2008

[Learning from minimal economic models](#)

T Grüne-Yanoff

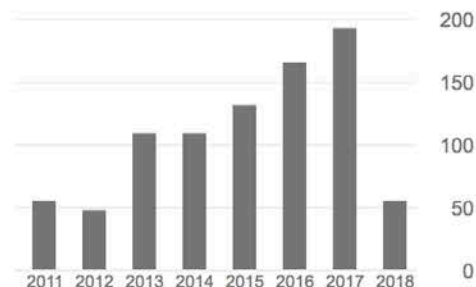
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2009

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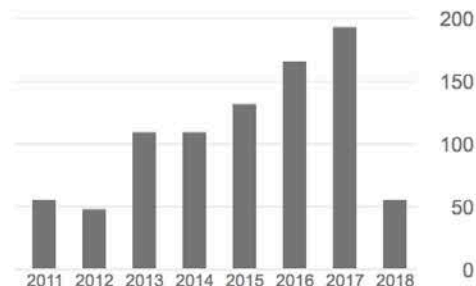
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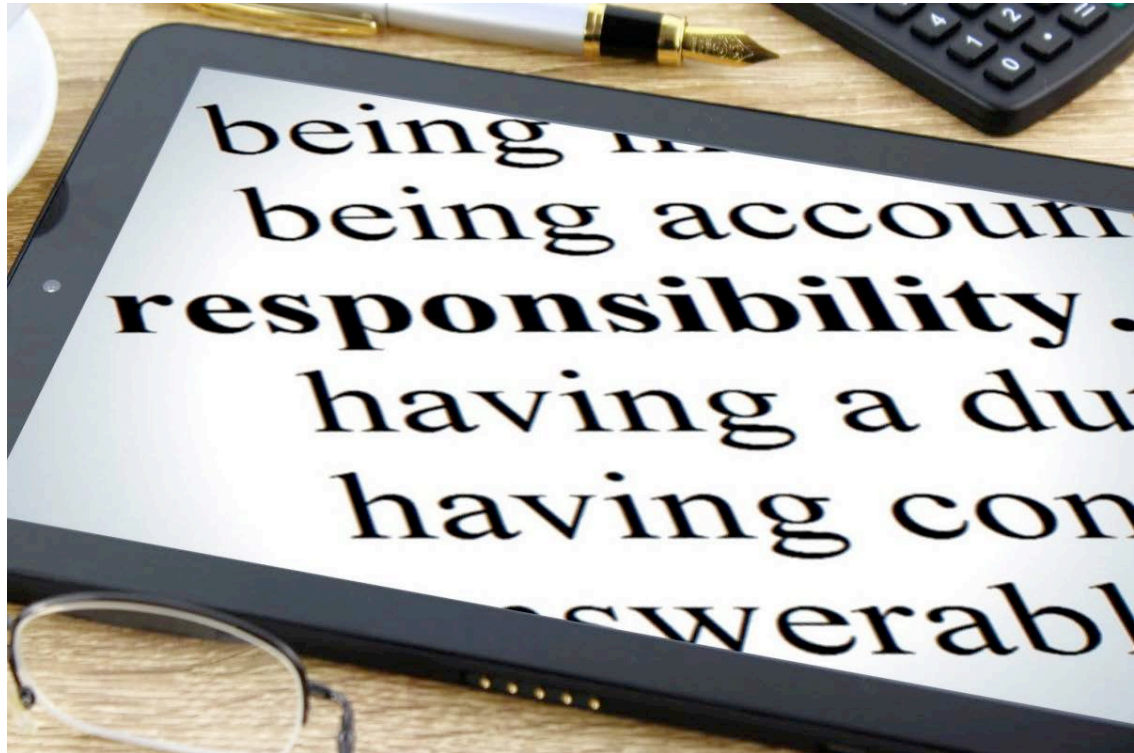
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|-----------|-----|-----|
| Citations | 973 | 769 |
| h-index   | 14  | 14  |
| i10-index | 23  | 19  |



**Publish or Perish**



Authorship establishes both credit and responsibilities





E. Eichinger (1892 – 1950) Der Gelehrte beim Schreiben in seinem



# Who is an author?

International Committee of Medical Journal Editors

.



# Who is an author?

International Committee of Medical Journal Editors

an *author* is involved in:

1. Substantial contributions to the conception of the work;  
AND
2. Drafting the work; AND
3. Final approval



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International Committee of Medical Journal Editors

an *author* is involved in:

1. Substantial contributions to the conception of the work;  
AND
2. Drafting the work; AND
3. Final approval; AND
4. Agreement to be accountable for all aspects of the work.



# Ethical Reasons for Authorship Criteria

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Consequentialism</b>                                                                                                                                         |
| <ul style="list-style-type: none"><li>• Impartiality of science</li><li>• Public trust</li><li>• Incentives for scientists</li><li>• Saving resources</li></ul> |

- Impartiality of science
- Public trust
- Incentives for scientists
- Saving resources



# Ethical Reasons for Authorship Criteria

| <b>Consequentialism</b>                                                                                                                                         | <b>Deontology</b>                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
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# Ethical Reasons for Authorship Criteria

| <b>Consequentialism</b>                                                                                                                                         | <b>Deontology</b>                                                                                       | <b>Virtue Ethics</b>                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Impartiality of science</li><li>• Public trust</li><li>• Incentives for scientists</li><li>• Saving resources</li></ul> | <ul style="list-style-type: none"><li>• Intellectual property rights</li><li>• Responsibility</li></ul> | <ul style="list-style-type: none"><li>• Sincerity</li><li>• Humility</li><li>• Resoluteness</li><li>• Reflexivity</li></ul> |



# Plagiarism

"The action or practice of taking someone else's work, idea, etc., and passing it off as one's own; literary theft"

(Oxford English Dictionary)





# Self-Plagiarism

The use of one's own previous work in another context without citing that it was used previously.





# Ghost Author

A person who writes a paper but is not listed among the authors





# Ghost Author

A person who writes a paper but is not listed among the authors



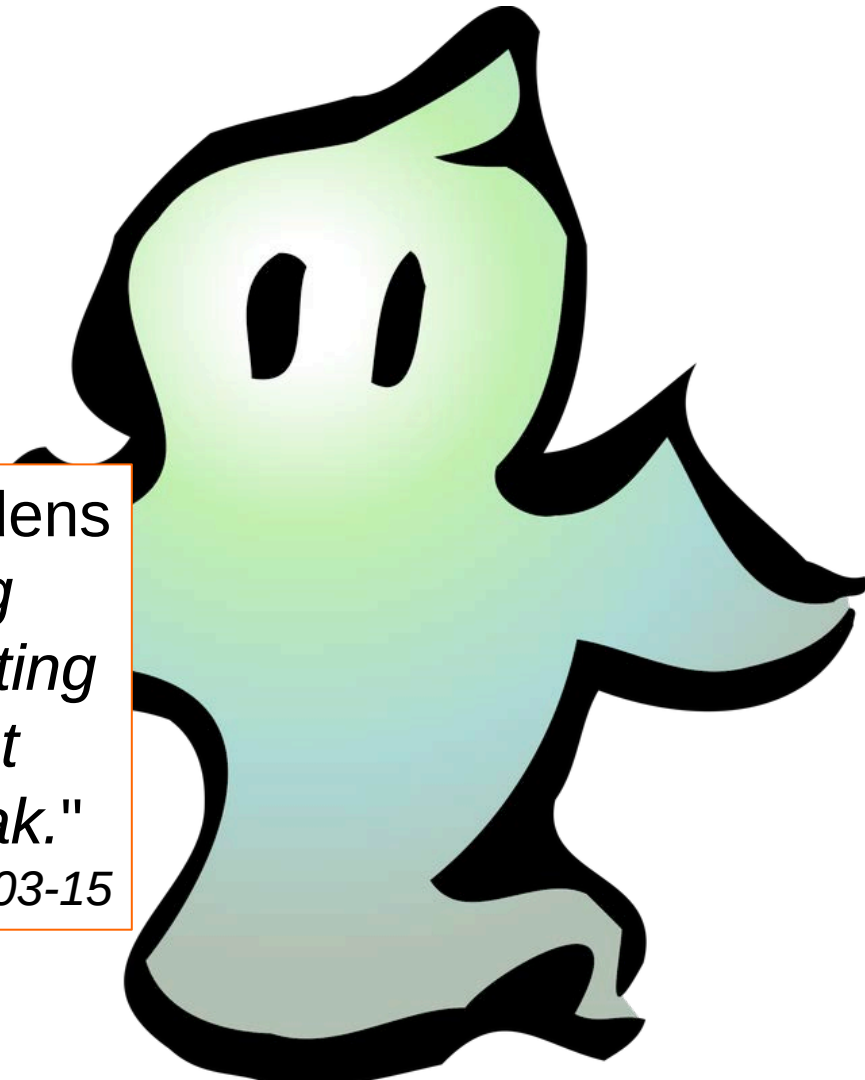


# Ghost Author

A person who writes a paper but is not listed among the authors

Monsanto executive William Heydens in an email: *"we would be keeping the cost down by us doing the writing and they [the scientists] would just edit & sign their names so to speak."*

*NPR, 2017-03-15*





Why is ghost authorship not permissible?  
Apply arguments from each of the three  
ethical frameworks.

Please pause the video and write down  
your answer before continuing!

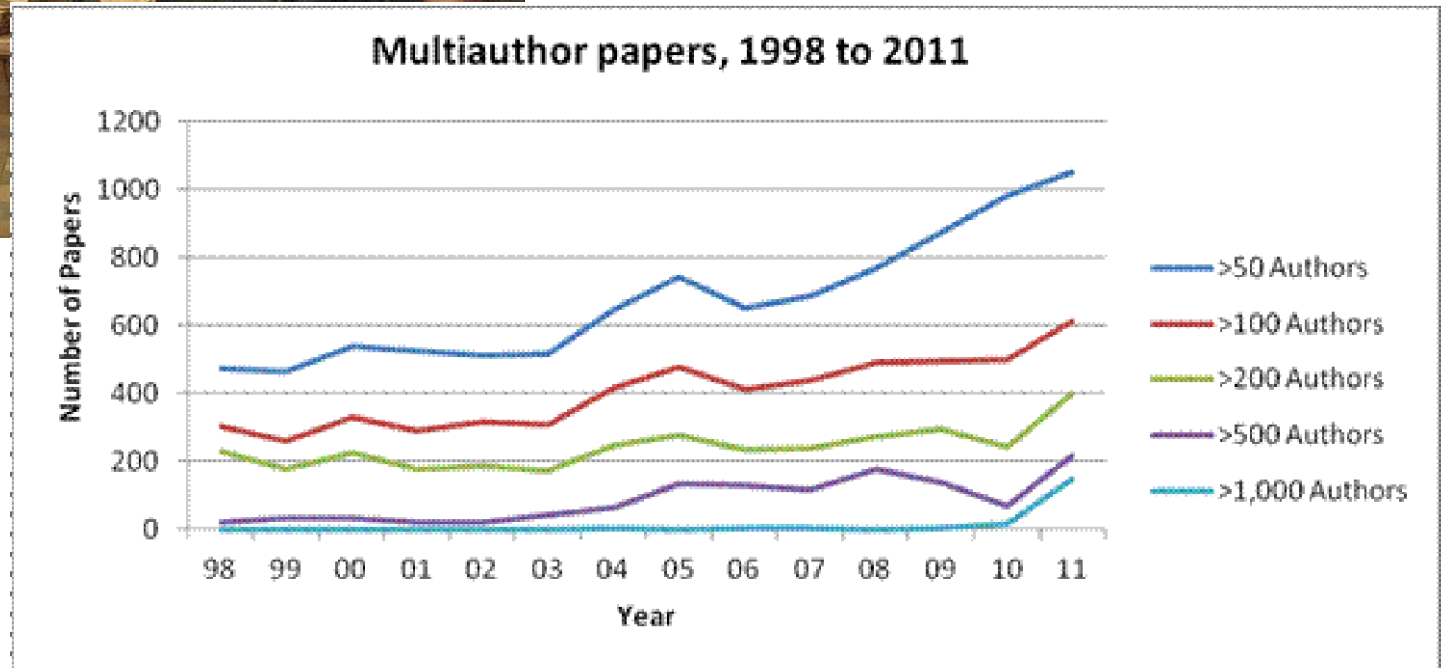
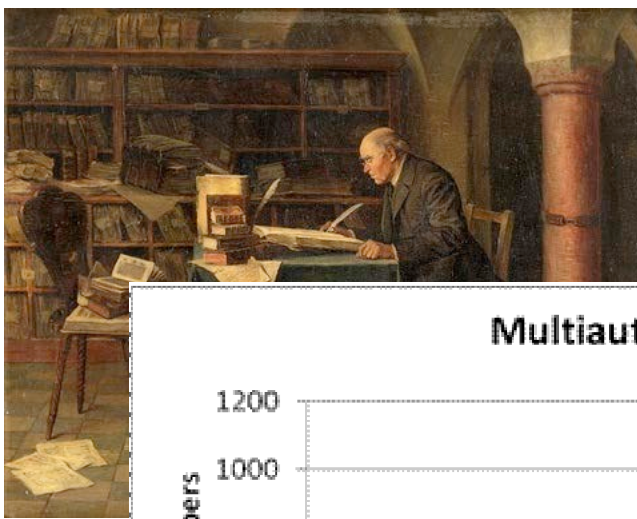


Why is ghost authorship not permissible?  
Apply arguments from each of the three  
ethical frameworks.

| Consequentialism                                                                                 | Deontology                                                       | Virtue Ethics                                                                        |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Impartiality of science</li><li>• Public trust</li></ul> | <ul style="list-style-type: none"><li>• Responsibility</li></ul> | <ul style="list-style-type: none"><li>• Resoluteness</li><li>• Reflexivity</li></ul> |



E. Eichinger (1892 – 1950) Der Gelehrte beim Schreiben in seinem



Source: ScienceWatch Newsletter, July 2012



# Honorary or Gift Authorship

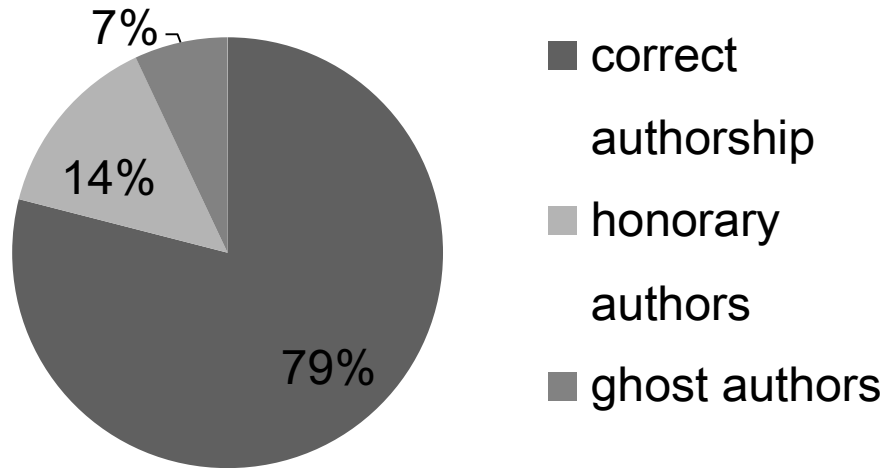
A person is presented as co-author but who does not satisfy the authorship criteria





# Honorary or Gift Authorship

A person is presented as co-author but who does not satisfy the authorship criteria



BMJ 2011 survey study: 630/896 responses





# Honorary or Gift Authorship

A person is presented as co-author but who does not satisfy the authorship criteria





# Honorary or Gift Authorship

A person is presented as co-author but who does not satisfy the authorship criteria

| Conceptual contribution | Non-conceptual contribution |
|-------------------------|-----------------------------|
|                         |                             |





# Honorary or Gift Authorship

A person is presented as co-author but who does not satisfy the authorship criteria

| Conceptual contribution | Non-conceptual contribution |
|-------------------------|-----------------------------|
| Study design            | Obtaining funds             |





# Honorary or Gift Authorship

A person is presented as co-author but who does not satisfy the authorship criteria

| Conceptual contribution      | Non-conceptual contribution |
|------------------------------|-----------------------------|
| Study design                 | Obtaining funds             |
| Data analysis decisions      | Commenting on drafts        |
| Data & result interpretation |                             |





# Honorary or Gift Authorship

A person is presented as co-author but who does not satisfy the authorship criteria

| Conceptual contribution      | Non-conceptual contribution |
|------------------------------|-----------------------------|
| Study design                 | Obtaining funds             |
| Data analysis decisions      | Commenting on drafts        |
| Data & result interpretation | Running experiments         |
|                              | Provide technical support   |
|                              | Collect data                |
|                              | Proofreading                |





# Order of Co-authors

Who is named as first, second, third,.... author?



PRL 114, 191803 (2015)

PHYSICAL REVIEW LETTERS

week ending  
15 MAY 2015

P. Bartos,<sup>144a,†</sup> A. Bassalat,<sup>117,‡</sup> A. Basye,<sup>165,‡</sup> R. L. Bates,<sup>53,‡</sup> S. J. Batista,<sup>158,‡</sup> J. R. Batley,<sup>28,‡</sup> M. Battaglia,<sup>137,‡</sup>  
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R. Beccherle,<sup>124a,124b,‡</sup> P. Bechtel,<sup>21,‡</sup> H. P. Beck,<sup>17a,‡</sup> K. Becker,<sup>120,‡</sup> M. Becker,<sup>43,‡</sup> S. Becker,<sup>100,‡</sup> M. Beckingham,<sup>170,‡</sup>  
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M. Begel,<sup>25,‡</sup> J. K. Behr,<sup>20,‡</sup> C. Belanger-Champagne,<sup>87,‡</sup> W. H. Bell,<sup>49,‡</sup> G. Bella,<sup>53,‡</sup> L. Bellagamba,<sup>20a,‡</sup> A. Bellierie,<sup>29,‡</sup>  
M. Bellomo,<sup>86,‡</sup> K. Belotskiy,<sup>98,‡</sup> O. Beltramello,<sup>30,‡</sup> O. Benary,<sup>153,‡</sup> D. Bencheikroun,<sup>156a,‡</sup> M. Bender,<sup>100,‡</sup>  
K. Bendtz,<sup>146a,146b,‡</sup> N. Berekos,<sup>10,‡</sup> Y. Benhammou,<sup>153,‡</sup> E. Benhar Noccioni,<sup>99,‡</sup> J. A. Benítez García,<sup>159b,‡</sup>  
D. P. Benjamin,<sup>45,‡</sup> J. R. Bensinger,<sup>23,‡</sup> S. Bentvelsen,<sup>107,‡</sup> L. Beresford,<sup>120,‡</sup> M. Beretta,<sup>47,‡</sup> D. Berge,<sup>107,‡</sup>  
E. Bergeas Kuutmann,<sup>106a,‡</sup> N. Berger,<sup>7,‡</sup> F. Berghaus,<sup>100,‡</sup> J. Beringer,<sup>15,‡</sup> C. Bernard,<sup>22,‡</sup> N. R. Bernard,<sup>76,‡</sup> C. Bernius,<sup>110,‡</sup>  
F. U. Bernlochner,<sup>71,‡</sup> P. Berry,<sup>77,‡</sup> P. Berta,<sup>129,‡</sup> C. Bertella,<sup>83,‡</sup> G. Bertoli,<sup>146a,146b,‡</sup> F. Bertolucci,<sup>124a,124b,‡</sup> C. Bertse,<sup>13,‡</sup>  
D. Bertse,<sup>113,‡</sup> M. I. Besana,<sup>91a,‡</sup> G. J. Besjes,<sup>108,‡</sup> O. Bessidskaia Bylund,<sup>146a,146b,‡</sup> M. Bessner,<sup>42,‡</sup> N. Besson,<sup>156,‡</sup>  
C. Betancourt,<sup>48,‡</sup> S. Bethke,<sup>101,‡</sup> A. J. Bevan,<sup>76,‡</sup> W. Bhimji,<sup>46,‡</sup> R. M. Bianchi,<sup>125,‡</sup> L. Bianchini,<sup>23,‡</sup> M. Bianco,<sup>30,‡</sup>  
O. Biebel,<sup>100,‡</sup> S. P. Bieniek,<sup>78,‡</sup> M. Biglietti,<sup>134a,‡</sup> J. Bilbao De Mendizabal,<sup>99,‡</sup> H. Bilokon,<sup>47,‡</sup> M. Bindl,<sup>54,‡</sup> S. Binet,<sup>117,‡</sup>  
A. Bingul,<sup>19c,‡</sup> C. Bini,<sup>132a,132b,‡</sup> C. W. Black,<sup>150,‡</sup> J. E. Black,<sup>143,‡</sup> K. M. Black,<sup>22,‡</sup> D. Blackburn,<sup>138,‡</sup> R. E. Blair,<sup>6,‡</sup>  
J.-B. Blanchard,<sup>136,‡</sup> J. E. Blanco,<sup>77,‡</sup> T. Blazej,<sup>144a,‡</sup> I. Bloch,<sup>42,‡</sup> C. Blocker,<sup>23,‡</sup> W. Blum,<sup>83a,‡</sup> U. Blumenschein,<sup>54,‡</sup>  
G. J. Bobbink,<sup>107,‡</sup> V. S. Bobrovnikov,<sup>109a,‡</sup> S. S. Bocchetta,<sup>81,‡</sup> A. Bocci,<sup>45,‡</sup> C. Bock,<sup>100,‡</sup> M. Boehler,<sup>48,‡</sup> J. A. Bogzaers,<sup>30,‡</sup>  
A. G. Bogdanchikov,<sup>109,‡</sup> C. Bohm,<sup>146a,‡</sup> V. Boisvert,<sup>77,‡</sup> T. Bold,<sup>38a,‡</sup> V. Boldea,<sup>28a,‡</sup> A. S. Boldyrev,<sup>99,‡</sup> M. Bomben,<sup>80,‡</sup>  
M. Bona,<sup>76,‡</sup> M. Boonekamp,<sup>136,‡</sup> A. Borisov,<sup>130,‡</sup> G. Borissow,<sup>72,‡</sup> S. Berroni,<sup>42,‡</sup> J. Bortfeldt,<sup>100,‡</sup> V. Bortolotto,<sup>60a,60b,‡</sup>  
K. Bos,<sup>107,‡</sup> D. Boscherini,<sup>20c,‡</sup> M. Bosman,<sup>12,‡</sup> J. Boudreau,<sup>123,‡</sup> J. Bouffard,<sup>2,‡</sup> E. V. Bouhova-Thacker,<sup>72,‡</sup>  
— — — — — 117 — — — — — 114 — — — — — 50 — — — — — 45 — — — — — 13 — — — — —



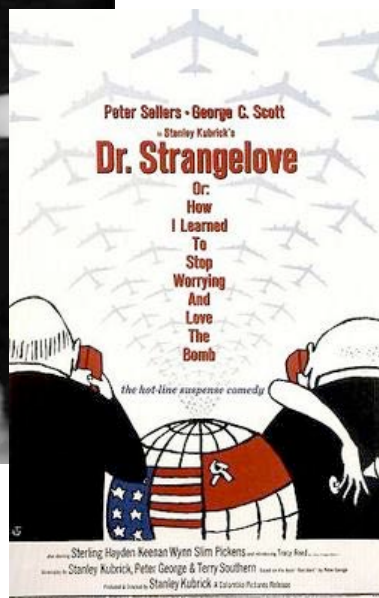
# Choosing Research Projects & Careers

# Dangerous Research Projects

---

## **Some research create risks:**

- Environmental dangers
- Health risks
- Economic, social or cultural harms



# Dangerous Research Projects

---

**Some research create risks**

What if the main goal is good but there are risky side-effects?

# Dangerous Research Projects

---

## Some research create risks

What if the main goal is good but there are risky side-effects?

*e.g. dual use technology*

“Dual use goods are products and technologies normally used for civilian purposes but which may have military applications.”



e.g. *dual use technology*

- Deontologists & virtue ethicists focus on individuals' intention or direct causal link – often unclear with dual use technologies
- Consequentialist : assess all possible consequences, irrespective of whether you intend them to happen or not

# Frivolous Research Projects

---

# Frivolous Research Projects

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## The IgNoble 2016

"Effect of Different Types of Textiles on Sexual Activity. Experimental study," Ahmed Shafik, *European Urology*, vol. 24, no. 3, 1993, pp. 375-80.

"Itch Relief by Mirror Scratching. A Psychophysical Study," Christoph Helmchen, Carina Palzer, Thomas F. Münte, Silke Anders, Andreas Sprenger, *PLoS ONE*, vol. 8, no 12, December 26, 2013.

# Frivolous Research Projects

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Palzer, Thomas F. Münte, Silk  
December 26, 2013.

## Biggest challenges of our time

- Food security
- Equality in economic growth
- The future of work
- Climate change
- Gender equality
- Future of healthcare

World economic forum 2016

## ***80,000 Hours* – A Consequentialist Perspective**

- some careers (research projects) aimed at doing good are far more effective than others
- You ought to choose that project...
  - ...with the highest potential for impact
  - ...which gives you individual better opportunities to have an impact in the future (e.g. through monetary gains)

For more, see <https://www.effectivealtruism.org/>

## Deontological counterargument



A human life is not just a means to produce outcomes, it is an end in itself. (Immanuel Kant)

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A human life is not just a means to produce outcomes, it is an end in itself. (Immanuel Kant)



Consequentialist framework does not allow showing partiality to one's own projects - a profoundly alienating and perhaps self-effacing moral theory (Bernard Williams)

# Conclusions

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- Methodological and research project choices raise many moral questions
- Many moral codes for scientists
- Focus on normatively valid justification of moral claims
- Three justificatory frameworks – all capture some aspects of our moral intuitions, but none seems to capture all
- Goal of this lecture: acquire concepts that help you in **your** reflection on specific problems – and to draw **your own** conclusions